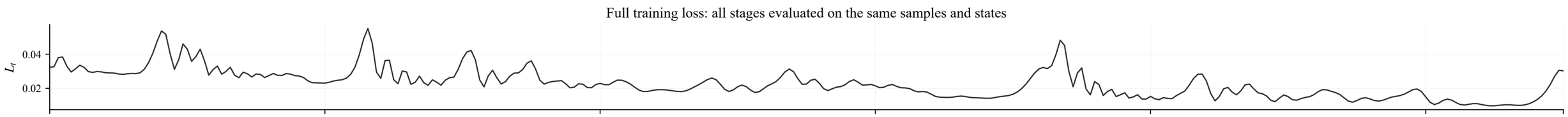
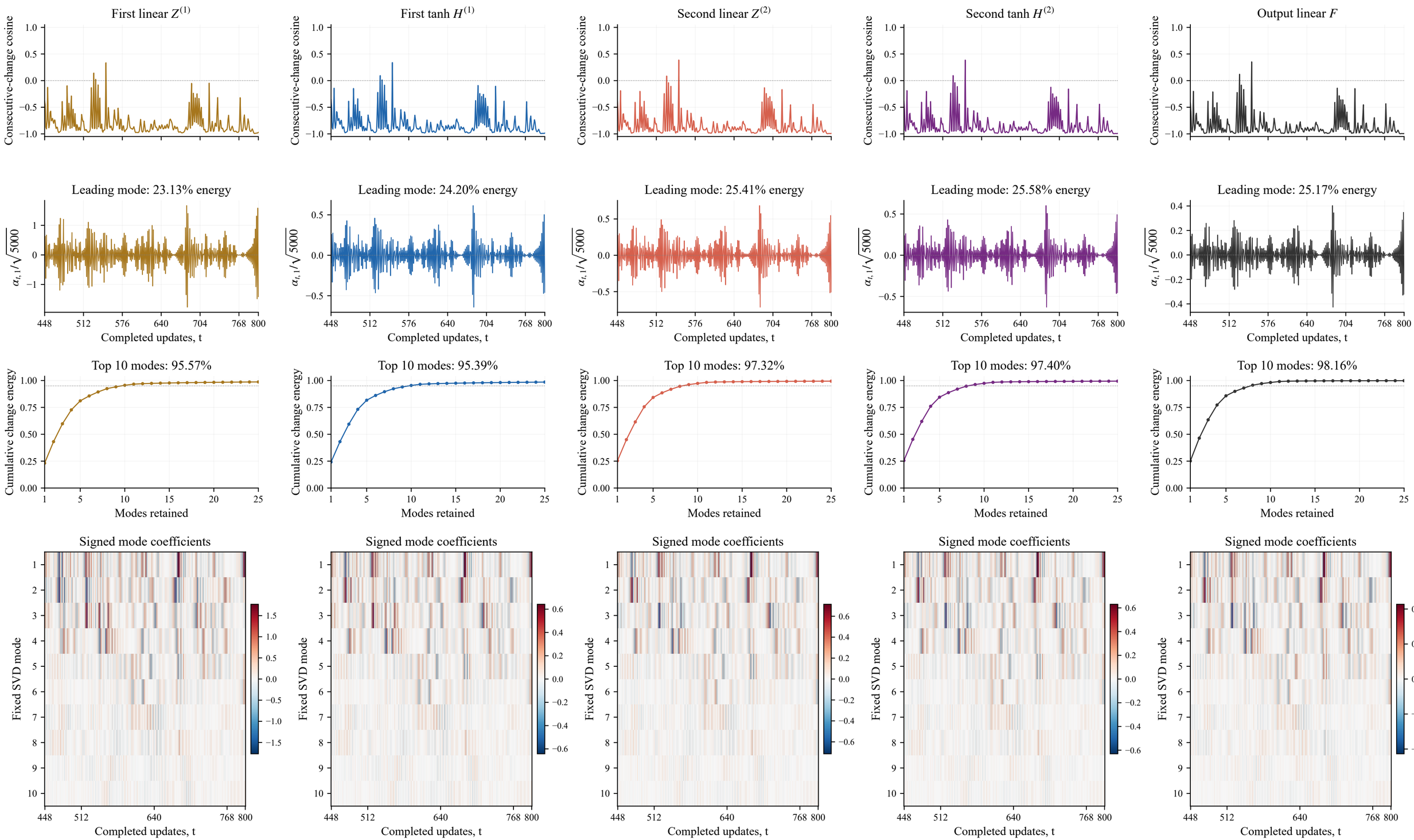
CIFAR-10 MLP: the complete forward path of the oscillation

Fixed input X (5,000 samples x 3,072 input coordinates) $\longrightarrow Z^{(1)} \longrightarrow H^{(1)} \longrightarrow Z^{(2)} \longrightarrow H^{(2)} \longrightarrow F$

INPUT CHECK: all 355 input matrices are exactly identical. $\Delta X = 0$; input-change rank = 0; cosine and normalized SVD fractions are undefined.





Each hidden/preactivation matrix has 5,000 sample rows and 200 activation columns; the output has 10 columns.

Each fitted row is an outgoing change $X_{t+1} - X_t$. No sample averaging, centering, state subsampling or per-mode color normalization.

Fixed SVD bases are fitted independently; equal mode numbers do not identify the same spatial directions across stages.